

# Jiwei Zhou

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## EDUCATION

**Purdue University, West Lafayette, IN**

Ph.D. in Technology

**May 2023**

GPA: 3.98/4.0

**Shanghai Jiao Tong University, Shanghai, China**

M.Eng. in Software Engineering

**June 2016**

**Zhuhai College of Jilin University (now Zhuhai College of Science and Technology), Guangdong, China**

B.E. in Mechanical Design & Manufacture and Automation

**July 2012**

## JOURNAL PUBLICATIONS

- J6. **Zhou, J.**, Camba, J. D., Company, P. (2026). CADialogue: A Multimodal LLM-Powered Conversational Assistant for Intuitive Parametric CAD Modeling. *Computer-Aided Design*, 191, 104006. <https://doi.org/10.1016/j.cad.2025.104006>
- J5. **Zhou, J.**, & Camba, J. D. (2025). The status, evolution, and future challenges of multimodal large language models (LLMs) in parametric CAD. *Expert Systems with Applications*, 282, 127520. <https://doi.org/10.1016/j.eswa.2025.127520>
- J4. **Zhou, J.**, Camba, J. D., Li, X. (2025). An Approach to Drawing Automation of Ship Stiffeners in the Shipbuilding Industry. *Computer-Aided Design and Applications*, 22(1), 26-41. <https://doi.org/10.14733/cadaps.2025.25-41>
- J3. **Zhou, J.**, & Hartman, N. W. (2024). A Model-Based Visual Inspection System (MBVIS) for Critical Plastic Bottle Dimensional Measurements. *Computer-Aided Design and Applications*, 21(2), 270-280. <https://doi.org/10.14733/cadaps.2024.270-280>
- J2. **Zhou, J.**, & Hartman, N. W. (2024). Development and Evaluation of a Vision Inspection System for Plastic Bottle Measurement. *Advances in Science and Technology*, 149, 41-50. <https://doi.org/10.4028/p-HPT9vc>
- J1. **Zhou, J.**, & Camba, J. D. (2021). Computer-aided process planning in immersive environments: A critical review. *Computers in Industry*, 133, 103547. <https://doi.org/10.1016/j.compind.2021.103547>

## PEER-REVIEWED CONFERENCE PROCEEDINGS

- C12. **Zhou, J.**, Camba, J. D. (*accepted*). EngDraw-Annotator: An AI-Assisted Interactive System for Evidence-Linked Engineering Drawing Annotation. *CIRPE 2026*.
- C11. **Zhou, J.**, Camba, J. D. (*accepted*). MAS-DFM: A Traceable Multi-Agent Framework for Drawing-Based Manufacturability Analysis. *2026 8th International Conference on Industry of the Future and Smart Manufacturing (ISM)*.
- C10. **Zhou, J.**, & Camba, J. D. (*presented*). EngDraw-Extractor: A Multi-Agent System for Information Extraction from 2D Engineering Drawings. *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2026)*.
- C9. Camba, J. D., Contero, M., and **Zhou, J.** (*accepted*). Large Language Models in Digital Manufacturing: An Empirical Study of Verification-Centered Design Cognition. *12th International Conference on Design Computing and Cognition (DCC'26)*.
- C8. **Zhou, J.**, Camba, J. D., Company, P., Contero, M. (*presented*). EngDraw-VQA: An Agent-Based Framework for Automated Visual Question Answering Generation From Engineering Drawings. *Manufacturing Science and Engineering Conference (MSEC 2026)*.

- C7. **Zhou, J.**, Camba, J. D., Company, P., Contero, M. Drawing-Checker: A Vision RAG Framework for Automated Comparison of Engineering Drawings. *Procedia CIRP*, 142, 127-132. <https://doi.org/10.1016/j.procir.2026.05.235>
- C6. Company, P., Camba, J. D., Contero, M., **Zhou, J.** (2026). From conceptual to embodiment design using Sketch-Based Modeling and Feature Recognition Techniques. *23rd annual International CAD Conference (CAD'26)*, <https://doi.org/10.14733/cadconfP.2026.50-54>
- C5. **Zhou, J.**, Gupta, D., & Camba, J. D. (2026). Prompt2CAD: A Lightweight LLM Framework for Conversational CAD Generation and Iterative Refinement. *Procedia Computer Science*, 277, 1663-1672. <https://doi.org/10.1016/j.procs.2026.02.204>
- C4. Gupta, D., Camba, J. D., Fuerst, T., & **Zhou, J.** (2025). WIP: An AI-Based Virtual Assistant for Supporting a Large Engineering Course. *IEEE Frontiers in Education Conference (FIE 2025)*, <https://doi.org/10.1109/FIE63693.2025.11328408>
- C3. **Zhou, J.**, & Hartman, N. W. (2023). A Framework for Model-Based Visual Inspection: A Case Study of Bottle Dimensional Measurements in the Plastics Industry. *2023 CAD Conference and Exposition*, 74-79. <https://doi.org/10.14733/cadconfP.2023.74-79>
- C2. **Zhou, J.**, Camba, J. D., Hartman, N. W., & Li, Z. (2022). An Approach to Extend the Digital Thread From Requirements to Model Geometry. *Manufacturing Science and Engineering Conference (MSEC) 2022*. <https://doi.org/10.1115/MSEC2022-80857>
- C1. **Zhou, J.**, Camba, J. D., & Fuerst, T. (2022). A Comparative Study on the Use and Interpretation of Annotated 3D Models. *IFIP 18th International Conference on Product Lifecycle Management*. [https://doi.org/10.1007/978-3-030-94399-8\\_23](https://doi.org/10.1007/978-3-030-94399-8_23)
  - 🏆 *Best Paper Award Nominee*

## **MANUSCRIPTS**

- M4. **Zhou, J.**, Camba, J. D. (*under review*). Human-AI Co-Design from Visual Requirements to Physical Validation: A Parametric Hydroponic Adapter Case Study.
- M4. **Zhou, J.**, Camba, J. D., Company, P. (*under review*). CADThread: A Multimodal Engineering Design Dataset Linking Feature-Based Parametric CAD, Geometry, and Drawings.
- M3. Gupta, D., Camba, J. D., Fuerst, T., **Zhou, J.** (*under review*). From Queries to Conversations: Exploring AI-Based Virtual Assistant Adoption, Usability, and Impact in Engineering Education.
- M1. Camba, J. D., **Zhou, J.** GenAI-Aided Design in Engineering Education: Competencies, Challenges, and Opportunities (Anticipated submission: 2026).

## **CONFERENCE PRESENTATIONS**

- P8. *EngDraw-Extractor: A Multi-Agent System for Information Extraction from 2D Engineering Drawings*. Oral presentation, **ASME 2026 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC-CIE 2026)**, Houston, Texas, USA.
- P7. *EngDraw-VQA: An Agent-Based Framework for Automated Visual Question Answering Generation from Engineering Drawings*. Oral presentation, **ASME 2026 International Manufacturing Science & Engineering Conference (MSEC 2026)**, hosted by Penn State University, State College, Pennsylvania, USA.
- P6. *Prompt2CAD: A Lightweight LLM Framework for Conversational CAD Generation and Iterative Refinement*. Remote presentation, **7th International Conference on Industry of the Future and Smart Manufacturing (ISM 2025)**, hosted by the University of Malta, Valletta Campus, Valletta, Malta.

- P5. *Integrating Artificial Intelligence in Plant Growth Monitoring for Innovative Rotary Aeroponic Systems*. Oral presentation, **2025 North American Plant Phenotyping Network Annual Conference (NAPPN 2025)**, hosted by the Donald Danforth Plant Science Center, Olivette, Missouri, USA.
- P4. *Development and Evaluation of a Vision Inspection System for Plastic Bottle Measurement*. Remote presentation, **10th International Conference on Mechanics, Materials and Manufacturing (ICMMM 2023)**, Washington, D.C., USA.
- P3. *A Framework for Model-Based Visual Inspection: A Case Study of Bottle Dimensional Measurements in the Plastics Industry*. Remote presentation, **20th International CAD Conference (CAD'23)**, Mexico City, Mexico.
- P2. *An Approach to Extend the Digital Thread from Requirements to Model Geometry*. Oral presentation, **ASME 2022 International Manufacturing Science & Engineering Conference (MSEC 2022)**, hosted by Purdue University, West Lafayette, Indiana, USA.
- P1. *A Comparative Study on the Use and Interpretation of Annotated 3D Models*. Remote presentation, **IFIP 18th International Conference on Product Lifecycle Management (PLM 2021)**, virtual conference.

## **PROFESSIONAL EXPERIENCE**

**Heliponix LLC, Evansville, IN**

**June 2023 – Present**

**Computer Vision Scientist**

- Conduct interdisciplinary research in applied AI to drive innovation in engineering design, smart manufacturing, and technology education.
- Lead the deployment of AI-powered vision systems in IoT-enabled smart appliances, enabling real-time crop monitoring and autonomous decision-making.
- Collaborate across hardware and software teams to design cyber-physical prototypes, integrating imaging sensors, edge AI, and cloud-based analytics for scalable smart gardening and precision agriculture solutions.
- Develop and optimize machine learning and deep learning algorithms, influencing the design of computer vision hardware and software.

**Digital Enterprise Center, Purdue University, West Lafayette, IN**

**August 2019 – May 2023**

**Graduate Research Assistant**

Drug Plastics - Model-based Metrology Project

**September 2021 – May 2023**

- Developed an innovative and cost-effective Model-Based Visual Inspection System (MBVIS) to enhance the critical dimensional measurement accuracy of plastic bottles in pharmaceutical applications, utilizing Geometric Dimensioning and Tolerancing (GD&T) information extracted from a Model-Based Definition (MBD) dataset, demonstrating high agreement with a commercial visual inspection system through Bland-Altman plot analysis
- Utilized Computer Vision techniques (OpenCV-Python and convolutional neural networks) to develop a more efficient process to measure critical bottle parameters

WHIN - Digital Product Information Project

**August 2020 – August 2021**

- Implemented RESTful APIs to facilitate interactive work instructions and Bill of Material (BOM) on a Product Data Management (PDM) system with Java
- Customized tooling management, process planning, workflow, project, and organizational structure on the PDM system to demonstrate to industrial partners

**Siemens Industry Software, Shanghai, China**

**July 2012 – July 2019**

**Application Engineer**

- Led testing and development of CAD software (Siemens NX), improving product reliability and reducing customer-reported issues by 15%.
- Collaborated with cross-functional teams to design and automate test workflows, achieving >95% code coverage and improved development efficiency.

- Supported partner universities through Siemens GO PLM initiatives, delivering CAD training and fostering academic–industry collaboration.
- Mentored junior engineers and coordinated quality assurance processes across multiple international projects.

**Siemens Industry Software, Shanghai, China**

**July 2011 – July 2012**

**NX Ship QA Intern**

- Tested on NX Ship Manufacture Module to improve ship product quality
- Created more than 300 reuse CAD parts to support NX Drawing Automation in the shipbuilding industry

## **FUNDING & GRANTS**

- **National Science Foundation (NSF) SBIR Phase II (Award No. 2151495)**—Partial support for research activities at Heliponix LLC
- **Project PID2022-137254OB-I00 (MCIN/AEI/FEDER, EU)**—Research support for peer-reviewed publications at Heliponix LLC
- **Wabash Heartland Innovation Network (Grant No. 4019008000 / 8000084103)**—Research support at Purdue University

## **PUBLIC SPEAKING & OUTREACH**

*Southwest Ecosystem Exchange*

**Evansville, IN**

**June 2025**

- Represented the company at the inaugural Southwest Ecosystem Exchange, sharing our startup journey and key support resources leveraged—including SBIR/STTR, match funding, and local partnerships—to inspire and connect with regional innovation leaders.

*Generative AI and Programming-Based CAD*

**Purdue University, West Lafayette, IN**

**Apr 2025**

- Delivered a 120-minute guest lecture covering programming-based CAD techniques, aligning with current research in Generative AI-enhanced design automation. Engaged graduate students through practical examples, interactive discussion, and Q&A.

*2025 CCO International Student Workshop – Internationally Friendly Employer Panel*

**Purdue University, West Lafayette, IN**

**Mar 2025**

- Served as a panelist sharing insights on transitioning from international graduate studies to careers in the U.S. tech industry, emphasizing the role of AI and computer vision in practical applications.

*Leveraging Artificial Intelligence for a Competitive Edge in Business*

**Indiana University, Bloomington, IN**

**Jan 2025**

- Delivered a keynote address on how businesses can leverage artificial intelligence for competitive advantage, highlighting opportunities, challenges, and strategies for adoption.

*Growing Smarter: How AI Helps Plants Thrive*

**University of Evansville, Evansville, IN**

**Jan 2025**

- Delivered a presentation on the role of artificial intelligence in enhancing plant growth and agriculture to inspire the next generation of innovators of the UE Explorers program.

*Journey from Academia to Industry*

**Purdue University in Indianapolis, Indianapolis, IN**

**Nov 2024**

- Presented a 30-minute talk to college and graduate students on transitioning from academia to industry, covering skill development, challenges, and career growth.

*Seeing with Smarts: How AI Helps Plants Grow!*

**Heliponix LLC, Evansville, IN**

**Nov 2024**

- Conducted an interactive session introducing core concepts of artificial intelligence and computer vision to middle and high school students, engaging them with practical applications in industry.

## **PROFESSIONAL SERVICE**

### **Technical Program Committee**

- International Conference on Mechanics, Materials and Manufacturing (ICMMM), 2026

### **Session Chair**

- ASME International Manufacturing Science and Engineering Conference (MSEC), 2026
- International Conference on Mechanics, Materials and Manufacturing (ICMMM), 2023

### **Conference Reviewer**

- ASEE Annual Conference & Exposition, 2023, 2024, 2026
- ASME International Manufacturing Science and Engineering Conference (MSEC), 2026
- ASME International Mechanical Engineering Congress and Exposition (IMECE), 2026
- IEEE Frontiers in Education Conference (FIE), 2025, 2026
- IEEE Visualization and Visual Analytics (VIS), 2026
- Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo, 2026
- International Conference on Industry of the Future and Smart Manufacturing (ISM), 2026

### **Journal Reviewer**

- *Applied Soft Computing*
- *Artificial Intelligence and Applications*
- *ASEE Computers in Education*
- *Computer-Aided Design*
- *Computers & Graphics*
- *Computers & Industrial Engineering*
- *Computers in Human Behavior: Artificial Humans*
- *Design Studies*
- *Engineering Applications of Artificial Intelligence*
- *Engineering with Computers*
- *Expert Systems with Applications*
- *Frontiers in Mechanical Engineering*
- *IEEE Transactions on Computational Social Systems*
- *Image and Vision Computing*
- *International Journal of Information Management Data Insights*
- *Results in Engineering*
- *SoftwareX*
- *Transactions on Graphics*

## **CERTIFICATION & TEACHING EXPERIENCE**

Siemens Certified NX CAD Instructor

Siemens Teamcenter Instructor, PLM Product Development Center (PPDC), Shanghai, China

- Offered formal training in CAD skills and Teamcenter application to new employees

**September 2011  
July 2013 – July 2019**

## **AWARDS AND SCHOLARSHIPS**

Conference Attendance Award

**Feb 2025**

- NAPPN Conference 2025 Supported by USDA NIFA DSFAS Program

Purdue Graduate Research Assistantship  
High-Performance Employee Awarded by Siemens PLM Software

August 2019 – May 2023  
2014 & 2016

## **LEADERSHIP EXPERIENCE**

Heliponix LLC, Evansville, IN

June 2023 – Present

*Research Associate Supervisor*

- Supervised a research associate, providing technical guidance and overseeing weekly progress on tasks related to plant science and computer vision

Christian Students at Purdue (CSaP), Purdue University, West Lafayette, IN

January 2020 – May 2023

*Vice President*

- Supervised the business of the organization and organized a weekly Bible study for club members

Siemens Industry Software, Shanghai, China

July 2012 – July 2019

*Mentor*

- Guided junior engineers and interns with professional CAD modeling and PLM application knowledge

## **MEDIA COVERAGE**

- *Southwest Ecosystem Exchange (2025)*. Presentation on Anu's startup journey and ESO partnerships. [\[Link\]](#)
- *Purdue University Center for Career Opportunities & Office of Professional Practice (2024)*. Invited panelist on international student career planning and transitions into U.S. industry roles. [\[Link\]](#)
- *Indiana University Kelley School of Business (2024)*. Invited keynote speaker for MBA consulting project on Food as Medicine, presenting Anu's AI-driven indoor farming approach to health and sustainability. [\[Link\]](#)
- *Purdue University (2022)*. Poster presentation on Realizing the Digital Enterprise projects, highlighted in university media coverage. [\[Link\]](#)